S(13): A COMPUTER-ASSISTED DETERMINATION

Theorem. S(13) = {2,3,5,7,11,13,17,19,23,29,31,37} = Primes(37).
Here the elliptic curve is arbitrary and the number field has exact
degree 13. The dyadic argument gives the global bound p <= 8374.

This proof uses exact modular-symbol, finite-field, and number-field computations. The proof inputs and their checking scripts are in the accompanying certificate packet. Failed preliminary computations are not proof inputs. The final coverage file has status PASS.

1. OCCURRENCE AND THE DEGREE CONDITION

The rational torsion primes 2,3,5,7 occur after extension to any field of degree 13. For each p=11,13,17,19,23,29,31,37, occurrences.json records an irreducible polynomial f of degree 13 and rational functions x=a, y=g(a)/h(a), where f(a)=0. Put

r=(x^2*y-x*y+y-1)/(x^2*y-x), s=(x*y-y+1)/(x*y),
b=r*s*(r-1), c=s*(r-1).

Then E: Y^2+(1-c)XY-bY=X^3-bX^2 has the point P=(0,0).
verify_occurrences.m proves that f is irreducible over Q, E is nonsingular, P is nonzero, and p*P=0, for all eight examples. Thus these are exact degree-13 occurrence certificates, not deductions from occurrence in a different degree.

The example at 37 is a van Hoeij degree-13 place. Its x-coordinate satisfies

x^13-4*x^12-x^11+10*x^10+3*x^9-21*x^8+3*x^7+24*x^6
-15*x^5-10*x^4+12*x^3-x^2-3*x+1=0.

Its y-coordinate, and the remaining examples, are recorded explicitly.

Conversely, a prime p dividing E(K)_tors is the order of a K-rational point. Its point on X_1(p) has degree dividing 13. For p>37 it cannot have degree 1, by the rational torsion theorem. Therefore it has degree 13. Let D be its irreducible rational trace divisor on X_1(p).

2. THE DYADIC GLOBAL BOUND

We specialize the previously proved dyadic argument, retaining its correction for the possible cotangent-lattice defect at 2.

Let J=J_0(p), T its integral Hecke algebra, e the winding element, I=Ann_T(e), and A=J/IJ the optimal winding quotient. Its rational points are finite. At 2 put

M=H^0(X_0(p)_{Z_2},Omega^1)[I], L=pi^*Cot(A_{Z_2}).

The integral coefficient pairing identifies M with the Z_2-dual of (T/I) tensor Z_2. Raynaud's theorem gives

$$2M \subset L \subset M.$$

We do not assume L=M or an optimal-quotient formal immersion at 2. The precise lattice argument and references are reproduced in the packet's earlier dyadic theorem.

Put m=26 and

T'_n=sum_{a|n} mu(n/a)T_a, L_n=T'_(2n)-2*T'_n.

The integral spans of L_1,...,L_m and (T_2-3)T_1,...,(T_2-3)T_m coincide. Moreover (T_2-3)Te lies in integral absolute homology. For the odd prime M0=157, DKSS's intersection formula, when p>2*m*M0=8164 and p*u=1 modulo M0, gives

L_n e . {0,a/M0}=epsilon_M0(na)-epsilon_M0(nu/a).

Here epsilon is the indicator of the upper half of the nonzero residues. The formula is an integral intersection formula and can therefore be reduced modulo 2.

For every u=1,...,156 the displayed 26 by 156 matrix has rank 26 modulo 2. dyadic_jets.json retains the rows; verify_dyadic.py rebuilds every entry and checks the rank using Sage. Consequently T_1e,...,T_26e are independent modulo 2Te: a dependence there, multiplied by T_2-3, would contradict the intersection matrices. Dualizing, the first 26 coefficient map on M is surjective. The cotangent image of
 f_26:X_0(p)^(26) -> A, Z |-> pi[Z-26*infinity]
at 26*infinity therefore contains 2*Z_2^26.

For clarity, the two dyadic local facts used next are elementary. If a map has cotangent image containing 2*Z_2^m, its formal pullbacks can be chosen as 2*x_i+Q_i(x), with Q_i integral of order at least 2. On coordinates in 4*Z_2, a nonzero vector of minimum valuation k>=2 cannot map to zero: one linear term has valuation k+1<2k.
Also, if an integral effective degree-d divisor reduces to d*c modulo 2, write its elementary symmetric polynomial as E_D(t)=1+2B(t). Then E_(2D)(t)=E_D(t)^2=1+4(B(t)+B(t)^2). Thus the doubled divisor agrees with 2d*c modulo 4, including ramified or repeated points. The torsion in the reduction kernel of an abelian scheme over Z_2 is killed by 2, since doubling maps its formal group into the torsion-free neighbourhood modulo 4.

Suppose now that p>(2^(13/2)+1)^2. At every place above 2, good reduction is excluded by Hasse’s bound. Additive reduction cannot support such prime torsion; nonsplit multiplicative reduction has odd torsion bounded by 2^f+1. In split multiplicative reduction the point must be in the component group, since p>2^f-1. Hence, after Atkin--Lehner on X_0(p), the trace divisor D0 of the image reduces to 13*infinity. The torsion point f_13(D0) reduces to zero, so f_26(2D0)=2*f_13(D0)=0. The doubled divisor lies in the neighbourhood of 26*infinity modulo 4. The preceding coefficient and local arguments force 2D0=26*infinity, a contradiction.

Since 8164 < floor((2^(13/2)+1)^2)=8374, this proves p<=8374. The use of 26 coefficients, rather than 13, is essential to this argument. No coarse polynomial constant from Parent is needed.

3. PUBLISHED EXCLUSIONS BELOW THE DYADIC BOUND

We use Derickx--Stoll, arXiv:2505.14109v1, Table 2 and Corollary 7.5, only within the table’s proved range p<100000. For p>=149 and p<=8374, its small-prime alternative fails because p^2>(2^19/325)*13+1. The CM alternatives require even degree. The other alternative would require (p-1)/(2*str(p)) to divide 13; no table entry or non-strange prime in this interval satisfies this condition.

For p=127,131,137,139, Proposition 4.1 applies with diamond <2>, ell=2,n=5. The gonality lower bound 325*(p^2-1)/65536 exceeds 65; Remark 4.4 gives the rational 2-torsion hypothesis. The respective diamond orders are 7,65,34,69. None of these orders, or any integral half or third of one, divides 13. Thus an irreducible degree-13 noncuspidal divisor cannot be a union of these orbits.

These integer checks exclude 1018 primes. The remaining 18 are

41,43,47,53,59,61,67,71,73,79,83,89,97,101,103,107,109,113.
global_bound.json records every prime and the published criterion
used for it.

4. ALL-RATIONAL-CUSP REDUCTIONS

We use the endomorphism version of the DKSS formal immersion test.
All maps are actual integral endomorphisms with finite rational image,
composed with torsion annihilators t_q=T_q-<q>-q, for odd q, in the
mu-level convention. They therefore kill the whole rational Jacobian
group. Fricke gives the equivalent test at the rational cusps in the
point-level model. This avoids any optimal-quotient lattice assumption
at 2.

For each possible rational-cusp reduction, the cotangent test consists
of independence modulo 2 of the indicated Hecke/diamond operator
prefixes. After normalizing a cusp of largest multiplicity to index
zero, it suffices to retain 13 coefficients there and 6 at every
other cusp. The selected intermediate diamond indices are
41:20, 43:21, 47:23, 53:13, 59:29, 61:15, 67:11, 71:5,
73:12, 79:39, 83:41, 89:11, 97:4, 101:5, 103:17,
107:53, 109:54, 113:56.
Several maps are combined at 41,43,61,67,73. At 43 the combined map
uses (<r>-1)t_q directly; at 107,109,113 the single such map suffices,
since these primes are non-strange. The other selected maps and
polynomials are explicitly generated by the FI exporters.

Every normalized multiplicity tuple of total degree 13 passes. For
small binary relation spaces the test enumerates all nonzero relations;
a relation with prefix lengths m0,m1,... can enter a normalized degree-13
tuple only if max(m0,m1,...)+sum_(i>0)m_i <=13. Otherwise the test
enumerates the prefixes themselves. At 79 all 2^25-1 relations were
tested in C++; Sage independently checks that the supplied relations
form a basis of the entire kernel. Other prefix counts are checked
against a separate dynamic-programming count. Compression is exact:
each original row is reconstructed from its compressed coordinates.
Sage independently verifies all 18 selected matrix ranks.
formal_immersion_final.json identifies the successful input files.

5. ORDINARY NONCUSPIDAL REDUCTIONS

census_sources.m enumerates every j-invariant over F_(2^f), f<=13,
all its twists, and all scalar classes of a point of order p. It
identifies closed points and full diamond orbits by their universal
Tate coordinates. The enumeration proves that every ordinary closed
point at the remaining levels has degree at least 7. Hence two such
points cannot occur in a degree-13 reduction.

In geometric point-level notation put A_3=T_3-1-3<3>. It kills rational
torsion and every rational cusp as an actual divisor. For each level
we use a finite-rational-image endomorphism alpha: alpha=1 at
41,47,59,71; alpha=sigma^6-1 at 61;
alpha=(sigma-1)*Phi_12(sigma) at 97;
alpha=(sigma-1)*(sigma^5+1) at 101;
alpha=sigma-1 at every other level. Here sigma generates the diamond
group. Table 2 and the published rank-zero results justify these maps.
Thus alpha*A_3 kills J(Q). An ordinary source P with only rational
cusps in its complement is impossible if alpha*A_3(P) is nonprincipal.

The full canonical F_2 models, integral at 2, are reused from the audited lower-degree work, with a new model at 113. Its genus is 477. The universal x,y units have pole degrees 167,175, verified at every cusp. The saturated q-expansion lattice has full rank modulo 2 through the Sturm bound. Global unit relations vanish to order greater than their section-degree bounds, and have rank g-1 at every source and Hecke target. This identifies actual canonical coordinates. The certified diamond acts by the inverse geometric point diamond.

stage2.m computes all degree-3 isogenies of each actual elliptic source and their weighted trace divisors, checks their total degree 4*deg(P), and constructs eight-term canonical jets. The ordinary local parameter is a6; the division-polynomial derivative is verified nonzero. Identical target support is identified before integral multiplicities and cancellations are formed.

Here is the elementary nonprincipality obstruction used by full_sieve.py. Write alpha*A_3(P)=U-V with disjoint effective U,V. Choose R=H^0(K^s), s=2 for p<=53 and s=1 otherwise; dim(R)>deg(U) is checked. Choose nonzero F in R(-U). If div(g)=U-V, then S=F/g is a nonzero member of R(-V), and H*S belongs to F*R for every H in R(-U). The stored homogeneous condition matrix on S has full rank, so such S does not exist. All multiplication tensors are verified by q-products beyond the appropriate Sturm bound. Each matrix retains its chosen F and H. Sage independently checks the final ranks.

There are 665 ordinary diamond orbits. The matrices exclude 663. The two remaining orbits are a degree-12 orbit of size 7 at p=43 and a degree-8 orbit of size 7 at p=113. The former is an actual lower-degree occurrence and is treated in Section 7. The latter is excluded by its diamond stabilizer below. Neither is asserted to have a nonprincipality certificate.

6. SUPERSINGULAR, CUSPIDAL, AND MIXED REDUCTIONS

The complete census gives the following supersingular degrees and numbers of closed points in a diamond orbit:
41:(5,2); 43:(7,3); 53:(13,1); 73:(6,4);
109:(9,3),(12,3); 113:(7,4).
The only nonrational cusp degrees at most 13 are
41:10, 43:7, 73:9, 89:11.
In addition to rational-cusp complements, the only mixed supersingular and ordinary supports are degree 5+8 at 41 and degree 6+7 at 73. The only repeated supersingular supports have degrees 5+5 at 41 or 6+6 at 73. These statements follow from the complete census and its minimum ordinary degrees. There are no omitted nonrational cusp mixtures.

At 41 the audited native function-field computations show A_3(P) nonprincipal for a supersingular degree-5 point, A_3(P+Q) nonprincipal for both unordered pairs up to diamond symmetry (including P=Q), and A_3(C) nonprincipal for a degree-10 cusp. They remain applicable with the respective rational-cusp complements of degrees 8,3,3.
At 43 the audited native supersingular degree-7 alpha*A_3 test applies with a rational-cusp complement of degree 6.

For the new mixed support at 41, sigma^2 fixes the degree-5 closed point. The matrices prove (sigma^2-1)A_3(P8) nonprincipal for both ordinary degree-8 source orbits. Since the same operator vanishes on the supersingular part, this excludes their sums. At 73 use

sigma^4-1, which fixes each supersingular degree-6 closed point; its A_3 test on the single ordinary degree-7 orbit is nonprincipal. These three additional obstruction matrices pass the independent Sage rank checks.

For the degree-7 nonrational cusp at 43 let C1,C2,C3 denote the three closed cusps. A primitive diamond cycles them. On this cusp type T_3(C)=<3>C+3C, so (<3>-1)A_3(C), up to diamond translate and sign, is 2*(2C1-C2-C3). cusp43.m proves this divisor nonprincipal by native Riemann--Roch. This excludes its sum with any rational-cusp divisor.

We use the following stabilizer argument for the other cases. Suppose h fixes the nonrational part of red_2(D), and let C be the rational lift of the rational-cusp complement. Put Z=D-C and y=[hZ-Z] in J(Q). Even if deg(Z) is nonzero, y is torsion when (h-1)J(Q) is finite: if h has order m, then

$$m*y=(h-1)[mZ-(1+h+...+h^{m-1})Z],$$

and the bracket has degree zero. Since y reduces to zero, 2y=0. If h has odd order and acts trivially on J(Q)[2], then (1+h+...+h^(m-1))y=my=0, whence y=0. The same conclusion holds when J(Q)[2]=0.

If y=0 and gon_Q(X)>13+deg(C), the principal divisor hD+C-D-hC is zero. If only 2y=0 is known, the bound gon_Q(X)>2*(13+deg(C)) suffices. In either case the disjoint cuspidal and noncuspidal supports give hD=D and hC=C separately. The rational cusps form a free diamond orbit, so m divides deg(C). We use gon_Q(X_1(p)) >= 325*(p^2-1)/65536 throughout.

At 73, h=<2> has order 9, fixes the degree-6 supersingular points and degree-9 cusps, and acts trivially on J(Q)[2] by the audited integral-homology computation on the actual JOne(73). The gonality bound exceeds 26. The cusp complements have degrees 7,1,4 for, respectively, one supersingular point, two supersingular points, and a degree-9 cusp. None is divisible by 9.

At 89, h=<2> has order 11. The gonality bound exceeds 39, hence exceeds 2*(13+2). A degree-11 cusp would require 11|2.

At 109, h=<23> has order 18 and fixes each degree-9 or degree-12 supersingular closed point. The gonality bound exceeds 58 and hence 2*(13+4) and 2*(13+1). The required divisibilities 18|4 or 18|1 fail.

At 113, h=<4> has order 7 and fixes the supersingular degree-7 closed points, as checked on their actual Tate coordinates. The gonality bound exceeds 63, hence 2*(13+6). This would require 7|6. The exceptional ordinary degree-8 orbit is fixed by h=<40>, of order 8. The same stabilizer test applies, since the gonality exceeds 2*(13+5)=36. Its rational-cusp complement would require 8|5. The Tate-coordinate stabilizer computation verifies this case as well.

At 53, h=<4> has order 13 and fixes the unique supersingular degree-13 closed point. Here C=0. A fresh computation on the integral sign-zero homology of JOne(53) shows that the common kernel modulo 2 of A_3 and A_5 is zero. Sage independently verifies these exact matrices. Thus J_1(53)(Q)[2]=0, so y=0. The gonality bound exceeds 13, giving hD=D. An order-13 diamond has no noncuspidal fixed points in characteristic zero: elliptic-curve automorphisms modulo sign have order at most 3. Thus D is one h-orbit of length 13. Its image on

X_0(53) is a noncuspidal rational point, contrary to Mazur’s rational prime-isogeny theorem. This completes the supersingular exclusions.

7. THE DEGREE-12 ORBIT AT 43

The degree-12 occurrence at 43 has x=a, where

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 f=t^12-t^11-5t^10+6t^9+9t^8-16t^7-3t^6+19t^5
   -3t^4-12t^3+t^2+4t+1,
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and

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 y=(35a^11-65a^10-107a^9+314a^8-34a^7-580a^6
    +601a^5+156a^4-552a^3+219a^2+136a-81)/43.
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The earlier exact number-field certificate verifies this point; its trace divisor B has degree 12. mixed43_lifting.m checks that f is irreducible modulo 2 and that its Tate coordinates reduce to a diamond translate of the exceptional closed orbit. Applying all diamonds therefore supplies a known rational degree-12 lift of each of its seven closed points.

The same script verifies formal immersion at P12+c for every rational cusp c, using the actual integral endomorphism whose cotangent matrix is AlphaA11Cotangent in the full canonical model. In q-eigenform coordinates its eigenvalue is

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 (chi(3)-1)*(a_11-chi(11)-11).
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In geometric point-level notation this is

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 (<3>^(-1)-1)*<11>^(-1)*(T_11-1-11<11>).
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It kills J(Q): the diamond difference has finite rational image and the Hecke factor kills rational torsion. Thus the argument uses an actual endomorphism, not an optimal quotient’s saturated cotangent.

Evaluation at P12 gives 12 independent binary cotangent columns; adding the q-coefficient column at any of the 21 rational cusps gives rank 13. All 21 matrices are retained and independently checked in Sage. Scaling the differential trivialization at P12 multiplies its 12 columns by an invertible binary matrix and does not change rank.

If an irreducible rational degree-13 divisor D had reduction P12+c, choose a diamond translate B’ of the known degree-12 trace lifting P12. Then D and B’+c reduce to the same divisor, and their degree-zero difference is killed by this endomorphism. Formal immersion implies the two integral points in this residue neighbourhood are equal. But B’+c is reducible. This contradicts the irreducibility of D.

8. COVERAGE, CHECKS, AND REFERENCES

The final coverage check accounts for every prime at most 8374: 12 occurrence primes, 1018 published exclusions, and 18 complete local exclusions. The dyadic proof excludes every larger prime. The exact ordinary obstruction matrices, mixed-support matrices, cotangent matrices, cusp checks, and number-field checks are all supplied.

The earlier canonical models and certificates at 41 and 43, and the 2-torsion action and stabilizer inputs at 73 and 109, were independently audited during the S(10)/S(11) work. The new degree-13 computations are checked here by the stated exact arithmetic verifiers. This does not claim a separate agent audit of the complete degree-13 proof.

Primary references:

- Derickx--Kamienny--Stein--Stoll, Torsion points on elliptic curves over number fields of small degree, Algebra Number Theory 17 (2023),

267--308; arXiv:1707.00364. In particular, its integral intersection formula and endomorphism formal-immersion criterion.
- Derickx--Stoll, Prime order torsion on elliptic curves over number fields, Part I: Asymptotics, arXiv:2505.14109v1. Proposition 4.1, Remark 4.4, Corollary 7.5, and Table 2 within its stated range.
- Mazur, Rational isogenies of prime degree, Invent. Math. 44 (1978), 129--162, for the nonexistence of a rational 53-isogeny.
- Van Hoeij, Low degree places on the modular curve X_1(N), arXiv:1202.4355, and the accompanying explicit low-degree-place data.
- The earlier dyadic theorem in asymptotic_prime_bound.tex, reproduced in this packet with its Raynaud, Abbes--Ullmo, and Manin-constant references. Its explicit coefficient test is replaced here by the 156 rank-26 matrices at M0=157.